



Regular Article

Empowering women through radio: Evidence from Occupied Japan[☆]Yoko Okuyama^{ID}

Kyrkogårdsgatan 10B, Department of Economics, Uppsala University, 753 13 Uppsala, Sweden

ARTICLE INFO

JEL classification:

A13, D72, J13, J16

Keywords:

Fertility

Political participation

Gender identity norm

Mass media

ABSTRACT

I study the impact of women's radio programs that the US-led occupying force aired nationwide in Occupied Japan (1945–1952) to dismantle the prewar patriarchal norms. From the perspective of the economics of identity, the radio messages can be viewed as attempts to alter gendered identity norms, and thus to shift women's political, economic and family outcomes. Using local variation in radio signal strength driven by soil conditions as an instrumental variable, I show that greater exposure to women's radio programs increased women's electoral turnout, and the vote share for female candidates, highlighting women's votes matter. I find no effects on women's labor market outcomes, but exposure to women's radio programs accelerated the postwar fertility transition. Overall, disseminating pro-gender-equality messages can have significant implications for both women's lives and society at large, potentially paving the way for rapid economic growth that would follow.

1. Introduction

Over the last century, societies have made a significant stride in expanding women's socioeconomic opportunities. Even wars and armed conflicts, albeit terrible ordeals, have provoked changes in the economic landscape, which can further serve as catalysts for advancing women's economic as well as political participation. However, a growing body of economic research suggests that preexisting gendered norms and attitudes often persist, continuing to hold women back. As highlighted in the reviews by Giuliano (2020) and Lowes (2022), much of the literature has focused on the historical persistence of these norms. What remains less understood is which public policies have weakened these constraints and driven changes in women's political, economic, and family outcomes.

In this study, I examine the impact of pro-gender-equality messages disseminated through radio on women's outcomes in Occupied Japan

(1945–1952). Beginning on October 1, 1945, the US-led occupying force (General Headquarters, Supreme Commander for the Allied Powers, or GHQ/SCAP) aired women's radio programs to challenge and dismantle deeply embedded patriarchal norms in prewar Japanese society (Japan Broadcasting Corporation, 1947). My textual analysis of GHQ/SCAP Weekly Radio Reports reveals that these programs promoted women's empowerment across three key dimensions: political participation, economic inclusion, and bodily autonomy. Drawing on Akerlof and Kranton (2000)'s economics of identity, I predict that these broadcasts served as a mechanism for reshaping gender norms, and increased women's electoral and labor force participation while accelerating fertility decline.

To empirically investigate the impacts of radio messaging, I build a unique district-level dataset that covers both the prewar and postwar periods. The dataset includes data on election results, labor force

[☆] This paper was previously circulated as “Toward Better Informed Decision-Making: The Impacts of a Mass Media Campaign on Women's Outcomes in Occupied Japan.” I am grateful to the editor and three anonymous referees for their helpful comments, and to my doctoral advisors Ebonya Washington, Costas Meghir, and Joseph Altonji for their guidance and support. I also thank Jason Abaluck, Jaime Arellano-Bover, Barbara Biasi, Cormac O'Dea, Alexia Delfino, Fabian Drixler, Olle Folke, Georg Graetz, John Eric Humphries, Mounir Karadja, Ilse Lindenlaub, John Marshall, Ken Miura, Chiaki Moriguchi, Emily Oster, Rohini Pande, Torsten Persson, Luca Repetto, Johanna Rickne, David Schönholzer, Oskar Nordström Skans, Luke Stein, John Tang, Kensuke Teshima, Junichi Yamasaki, Jessica Vechbanyongratana, and seminar participants at various seminars including the Labor/Public Prospectus Workshop at Yale University, IIES Stockholm, Stockholm School of Economics, Uppsala University, the Norwegian School of Economics, Colgate University, the Online Seminar on Gender Economics, the Kobe Development Economics and Economic History Seminar, the National Graduate Institute for Policy Studies (GRIPS), Hitotsubashi University, ASSA 2021 Virtual Annual Meeting, Virtual Economics History Seminar, University of Michigan Center for Japanese Studies, and the University of California Berkeley. In addition, I would like to thank Haruko Nakamura at the Yale East Asia Library for providing access to prefectural yearbooks, Miriam Olivares at the Yale Center for Science and Social Science for ArcGIS support, the StatLab and the Digital Humanity Lab at Yale University Library for their consultative support on map digitization, Kazuo Kishimoto for sharing with me the digitized Population Census data and the House of Representatives election results, and Ayumi Sudo for excellent research assistance. Finally, I acknowledge funding for digitizing historical materials from the Cowles Foundation. All mistakes are my own.

E-mail address: yoko.okuyama@nek.uu.se.

<https://doi.org/10.1016/j.jdevec.2025.103620>

Received 13 September 2024; Received in revised form 19 August 2025; Accepted 22 August 2025

Available online 6 September 2025

0304-3878/© 2025 The Author. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

participation, marriages, births, and the geographical reach of the radio, gathered from various historical resources such as the Japan Broadcasting Corporation's reports, maps, newspapers, official election results, newly digitized vital statistics, and population census. Exposure to women's radio programs, which is my main independent variable, comes from district-level variation in radio listenership. Listenership is proxied by the district-level radio subscription rate, a statistics that originates from the state-owned radio licensing system at the time. Across the nation, listenership varied from below 10 percent to above 70 percent with an average of 37 percent. This provides cross-sectional variation in women's exposure to messages regarding gender-egalitarian norms on air.

A key empirical challenge is that radio listenership may not have been randomly assigned. Areas with higher listenership could have differed systematically in ways that influence women's outcomes. To address this potential endogeneity, I employ an instrumental variable (IV) strategy that exploits variation in AM radio signal strength driven by local soil conductivity. Signal quality is generally determined by factors such as distance to a transmitter and transmission power, and I control for these directly. I then use the remaining variation in field strength, attributable to soil characteristics, as the instrument for radio listenership. This approach isolates plausibly exogenous variation in exposure to women's radio programs. The identifying assumption is that soil-driven differences in radio signal affect women's political participation, labor market activity, and fertility only through increased exposure to radio programs, and are not correlated with unobserved determinants of these outcomes. I support this assumption by showing that the IV is uncorrelated with a range of preexisting local characteristics, including measures of female empowerment, democratic penetration, attitudes toward the wartime military government, sectoral composition, and pre-U.S. occupation outcomes of interests.

My findings are three-fold. First, I find that greater exposure to women's programs significantly increases women's voter participation in the first postwar election where women were allowed to vote. Specifically, a one standard deviation increase in exposure to women's radio programs increases women's electoral turnout by 2.4 percentage points. Based on this estimate, the persuasion rate (DellaVigna and Gentzkow, 2010) is 18.1, which has a comparable size with, but larger than, the persuasion rates of mass media on voter participation in historical contexts (Gentzkow et al., 2011), and close to the rate achieved by effective Get-Out-the-Vote (GOTV) campaigns.

In areas with greater radio exposure, not only female voter participation, but also vote share for female candidates was greater: a one standard deviation increase in exposure also raises a female candidate's vote share by 2.3 percentage points. This positive effect contributes to women's increasing representation in the national legislature. By calculating the counterfactual election outcome with zero radio exposure in every district, I demonstrate that women's share among winners would have been only 5.3 percent, compared to the actual representation of 8.2 percent. This finding makes a case against the *double-vote hypothesis*, which posited that women, especially married women, would simply cast the same votes as their husbands and have no influence on electoral outcomes. My finding, by contrast, highlights that women's votes matter.

Second, I find that radio exposure *accelerates* the decline in the number of births. A one standard deviation increase in radio exposure contributes to decreasing the 1950 birth rate by 2.0 per 1000 population. Around that time, Japan's birth rate began to decline as it entered a fertility transition,¹ following a brief postwar baby boom. My findings imply that radio messages contributed to speed up the

fertility transition. Data show that the present effect is not likely due to increased abortions. I also provide supporting evidence that the decline in fertility is not simply a delay in births, but likely a decrease in the number of lifetime births per woman. Overall, my results imply that the empowerment of women contributed to the notably rapid fertility transition in postwar Japan.

Third, one aspect unaffected by the radio messages was women's participation in the labor market. This could be due to a tightened labor market following the influx of men returning from World War II. It could also be due to the persistent traditional attitudes of male-dominated employers and men's view on working women remained unchanged. This is plausible, considering that the radio campaigns were primarily directed at women, possibly overlooking the need to simultaneously change men's norms.

Further analyses suggest that the main findings are likely driven by empowering messages directed at women rather than by general radio exposure. Moreover, the impacts do not vary by women's education level or by whether districts are urban or rural, both factors that would typically reflect informational gaps. This indicates that information alone is not the primary channel, and instead the results point to the role of normative messages. To corroborate this interpretation, I provide correlational evidence showing that women's higher postwar radio listenership is associated with more egalitarian gender norms. Finally, the radio impacts are stronger in areas with lower male to female ratios, suggesting that a greater presence of women in the locality amplifies the effects.

Contribution to literature. This study contributes to three strands of literature. First, this study provides evidence that mass media can help disseminate messages that women's voice counts in elections, which get them out to vote and influence the electoral outcomes. Generally, the literature on mass media and electoral turnout has shown that political information disseminated through mass media increases voter turnout (DellaVigna and Gentzkow, 2010; Gentzkow et al., 2011; Strömberg, 2004, 2015). This line of literature, however, mostly focuses on long-enfranchised voters. Meanwhile, my study shed light on the newly enfranchised women. They deserve careful attention. Previous studies suggest that "[o]verall turnout declined as a result of adding women to the eligible electorate". (Corder and Wolbrecht, 2016). While data on gender-specific turnout are rarely available, a few studies confirm that women turned out less than men did (Corder and Wolbrecht, 2016) when they were newly franchised across the globe. That is to say, women's suffrage, albeit a crucial milestone in advancing the position of women, may not automatically translate into equal exercise of political power between men and women. My findings highlight that disseminating new norms through mass media can be a way to move the needle: it mobilizes more women into the active electorate and influences electoral outcomes. Mass media help to incorporate women into the electorate.

Second, this study contributes to the economic literature on the impact of women's empowerment on fertility, particularly in settings where fertility rates remain high. Prior studies have shown that women's empowerment influences fertility through four economic channels: changing women's *opportunity cost* of time, enabling access to (Goldin and Katz, 2002; Bailey, 2006) and information (Kearney and Levine, 2015) of contraceptives and reproductive technologies, enhancing women's relative *bargaining power* (Doepke and Kindermann, 2019), and altering *public attitudes and norms* (see Silva and Tenreiro (2017) for review). The present study relates to the norm-transmission channel but differs in key ways from previous work. The radio programs I study directly targeted women and promoted birth control to improve women's health and empowerment. This contrasts with the dissemination of family planning through court rulings reported in newspapers studied by Beach and Hanlon (2022). While La Ferrara et al. (2012) shows that commercial entertainment television portrayed small families as role models and reduced fertility, the programs I study

¹ In 1947, the crude birth rate was 34.5 per 1000 population and the total fertility rate was 4.54. By 1955, the crude birth rate had declined to 19.5 per 1000 and the total fertility rate to 2.37, according to the National Institute of Population and Social Security Research.

were publicly run by the occupation authority, making the findings more directly policy relevant. Whereas Jensen and Oster (2009) find larger effects of cable television in rural India with limited access to information, I do not find urban-rural differences, suggesting that information is not the likely channel. Finally, I find no evidence that radio exposure increased stillbirths, a proxy for abortions, which indicates limited unintended consequences. This is an additional aspect not previously examined in the literature.

Third, this study speaks to the economic literature that examines the impacts of radio broadcasting. Studies show that radio messages work by reducing information friction (Strömberg, 2004; Svensson and Yanagizawa-Drott, 2009; Russo, 2019), disseminating propaganda or populist persuasion (Yanagizawa-Drott, 2014; DellaVigna et al., 2014; Adena et al., 2015; Wang, 2021a), changing time use (Olken, 2009), coordinating social movement (Gagliarducci et al., 2020; Wang, 2021b), or facilitating norms and attitudinal changes (Cheung, 2012; Blouin and Mukand, 2019; Khalifa, 2022). While the mechanism at work depends on the context, the norm channels appear to fit the present study. In terms of empirical strategies, this paper follows the spirit of Strömberg (2004) and Adena et al. (2015), which identify the impact of AM radio waves using ground conductivity as an instrumental variable. The empirical context is also unique, as the radio programs in this study were broadcast by a foreign occupier and implemented on a large scale. Additionally, this study provides a quantitative content analysis, which has been rarely undertaken in the existing literature due to limited data availability.

Organization of the paper. The remainder of the paper is organized as follows. Section 2 provides a brief background of women's radio programs in Occupied Japan and highlights key features that are critical for my empirical analysis. Section 3 presents the hypothesis. Section 4 describes the data sources, and Section 5 outlines the identification and estimation strategy. Section 6 reports the empirical results. Section 7 considers the robustness of the findings, potential mechanisms, and the generalizability of the results. Section 8 concludes.

2. Historical background

This section provides three elements of historical context essential to the study. Section 2.1 discusses how women's radio programs were among the Allies' first initiatives to raise women's social status. Section 2.2 describes the preexisting radio broadcasting infrastructure, which underpins the identification strategy. Section 2.3 analyzes the content of women's radio programs, both qualitatively and quantitatively, and examines women's political participation, labor market activity, and family formation as relevant behavioral outcomes.

2.1. The allies' efforts to raise women's status in occupied Japan

In prewar Japan, women lived under the influence of patriarchal norms. At school, they were taught the ideology of "Good Wife, Wise Mother (*Ryosai Kenbo*)". This concept refers to the ideal image of a woman at the time, rooted in the belief that a woman's primary duties are to manage the household, bear children, and raise them.² Meanwhile, women were excluded from politics — they were not only denied voting rights but were also prohibited from joining political organizations.³ Upon marriage, women were considered "incapacitated".

² Studies show that textbooks for girls approved by the Ministry of Education adopted the ideology of "Good Wife, Wise Mother (*Ryosai Kenbo*)" (Koyama, 1991; Jiang, 2022).

³ The Constitution of the Empire of Japan (*Meiji Constitution*), enacted in 1890, did not grant women voting rights. Moreover, the Association and Political Meetings Law of 1890 (*Shukai Oyobi Seisya Ho*) and the Article 5 of Public Safety and Police Law of 1900 (*Chian Keisatsu Ho*) barred women from joining a political organization or even just attending a political meeting. In

They needed their husband's permission to work, and their property was controlled by their husbands.⁴ Their pursuit of birth control has also encountered resistance. The birth control movement began to gain momentum during the 1920s,⁵ but the movement was suppressed when Japan entered the war in the 1930s. At that time, the complete fertility rate was 4.6 per woman. During the war, the Japanese government implemented pro-natal policies to raise this number to over five. Marriages were encouraged by municipalities, married women without children were inspected, and parents with more than ten children were rewarded (Ogino, 2008).⁶

After World War II, the Allied Occupation⁷ prompted a seismic shift in women's status in Japan. The occupation's five major reform directives included raising the legal and social status of Japanese women.⁸ GHQ/SCAP attributed prewar militancy to Japan's patriarchal system (Kobayashi, 2004), and prioritized raising women's status as a key peacebuilding strategy. During the occupation period, Japanese women gained new legal rights, including rights to vote and run for office (in December 1945 for the national election, and in September 1946 for the local election), and greater access to higher education (1948). The new constitution, enacted in May 1947, has prohibited discrimination on the basis of sex.⁹ Indeed, the U.S. occupying force "chose to make Japan a laboratory for one of the world's most radical experiments with women's rights" (Pharr, 1987).

GHQ/SCAP viewed a change in women's mindset as crucial for raising women's status and, by extension, for the overall success of the Allied occupation. Therefore, their initial initiative was to launch radio programs targeting women. In fact, as early as October 1, 1945, just one month after the Allies started to occupy Japan, the government-sponsored radio station, the Japan Broadcasting Corporation, began to air programs targeting women. The women's programs aimed "to raise political, social, and cultural standards of ordinary women and breaking away from feudalism", and "in order to select qualified female leaders, [the women's radio programs introduced] not only anti-militarists who remained silent during the war but also many unknown progressive, young women" (Japan Broadcasting Corporation, 1947; translated by the author).

1922, the Article 5 of Public Safety and Police Law was amended in response to women's petition and women were allowed to attend a meeting. However, they continued to be barred from joining any political organization. The women's suffrage movement also gained momentum in the 1920s, yet it did not achieve success in the prewar period.

⁴ The Articles 14–18 of the Civil Code of 1898 (*Meiji Minpo*).

⁵ Baron Shizue Ishimoto (later Shizue Kato) started the birth control movement after she met an American birth control advocate, Margaret Sanger, in New York. However, this movement encountered resistance in the late 1930s.

⁶ The Ministry of Welfare introduced the award (*Yuryo Tashi Katei Hyosho*) in 1940. In its first year, 10,336 parents received the award. The following year, an additional 2134 parents were recognized.

⁷ After World War II, Japan was occupied by the Allied Powers from September 2, 1945 to April 28, 1952. Although officially called the "Allied Occupation", it was mostly an undertaking of the United States, with contributions from Australia, India, New Zealand, and the United Kingdom, and therefore it was often called the "American Occupation". General Douglas MacArthur oversaw the occupation as the Supreme Commander for the Allied Powers (SCAP). The acronym SCAP was soon used to refer not only to the commander himself, but also to the offices of occupation set up under him to guide Japan to demilitarize and democratize the nation.

⁸ The other four reforms were to abolish the secret police, to encourage the formation of labor unions, liberalize the education system, and democratize the economy. Source: Diplomatic Records A 1.0.0.2-3-4 "Conference Abstracts and Memoranda between the Supreme Commander for the Allied Powers and his Staff and the Prime Minister and other ministers of Japan" <GAI-1, Reel No. A'-0055>.

⁹ Uemura (2007)'s pioneering work uncovers the role of the Occupation regime in raising Japanese women's status and, in particular, analyzes the relationships between occupier and occupied.

2.2. Radio reception and use in occupied Japan

At the onset of the Allied Occupation, there were 53 radio transmitters and 39 amplifiers across the nation, all of which were connected and operated by a single state-sponsored radio station, the Japan Broadcasting Corporation (JBC). The JBC has a primary channel, channel 1 (*daiichi hoso* in Japanese) which aired various programs throughout the day, and a secondary channel, channel 2 (*daini hoso*) which was used for only part of the day and mainly for rebroadcasting. Until 1952, there was no private radio broadcasting. In effect, a radio holder had only two choices: either to listen to the JBC's programs, or not to listen to any programs at all. Such a binary choice set turns out to be critical for my empirical analysis. I do not need to consider the listener's selection into different radio stations at the same time window.

Importantly for my analysis, the JBC kept records on the number of households subscribing to radio as well as the total number of households in all municipalities. This is because the JBC mandated all radio holders to register and pay subscription fees. The fee was not expensive and thus, I do not worry that it may have excluded low-income families from acquiring radio. In fact, the mandated annual fee in 1950 was 35 yen, which was 2.5 times the price of one serving of Soba noodles, Japanese staples.¹⁰ I digitize the JBC record to calculate the radio subscription rate, which I later use as an independent variable in my empirical analysis.

During the Allied Occupation, the JBC operated under the close supervision of the GHQ/SCAP Civil Information and Education Section Radio Unit (later also called the Radio Branch and hereafter referred to as the Radio Unit). Radio broadcast content was censored in advance by the Radio Unit.¹¹ The Radio Unit also conducted modern listener surveys (Mayo, 1988; Luther and Boyd, 1997; Smulyan, 2002). In effect, the Radio Unit had a large say over what kind of content was on air and therefore played a key role in disseminating information to meet the GHQ's purposes.

The JBC started airing the flagship women's program "Women's Hour (*Fujin no jikan*)" on October 1, 1945, just about a month after the Allies started occupying Japan. The program was designed to draw as many women as possible. A time slot allotted to the women's program was the lunch break when women used to listen to a war-time women's program during World War II; Japanese women were appointed as the director of the production team, as well as moderators, in order to be friendly to female listeners; music was played occasionally for a pause so that listeners could maintain concentration (Japan Broadcasting Corporation, 1947, 1950).

As the JBC's Listener survey reveals, women's programs were indeed well received. In the 1947 survey, more than 70 percent of women with a radio subscription said that they currently listened to or used to listen to the women's program. Not only did they listen to the women's program, but more than 60 percent of them answered that they had gained new insights through the program.

As time went by and the JBC's production capacity increased, the JBC added more time slots for women's programs. By the end of 1950, the weekly airtime that the JBC allocated to women's programs had quadrupled from its onset in October 1945.¹² This underscores the fact

that GHQ/SCAP maintained and strengthened efforts to raise women's status throughout the occupation. As the airtime expanded, the content covered by the women's programs grew too, as I show in detail in the next subsection.

Before diving into the radio content analysis, I note that the analysis primarily focuses on the occupation period (1945–1952) although the JBC continued to air women's programs until 1963. I restrict my attention mainly to the occupation period because, at the end of the occupation period, private radio broadcasting as well as TV broadcasting started, giving more choices to potential listeners. Competition among different mass media outlets may have fundamentally changed the nature of media content as well as complicated the listeners' decision process about which information they acquire and why. Although this transition in the broadcasting market opens up a new avenue of research, it is beyond the scope of my current analysis.

2.3. Content of women's radio programs

What kind of messages did the women's programs try to disseminate? Answering this question is key to determine which women's outcome this study should consider. Therefore I turn to the Weekly Radio Reports (from January 1946 to December 1950), which document daily radio content. In the following, I first present some examples. Then, I classify the contents systematically into several topics and evaluate how the composition of these topics changed throughout the duration of the Allied Occupation.

The Weekly Radio Report¹³ was reported every week in English, with one section dedicated to the featured programs of the week. The first example below sheds light on a significant change in norms concerning birth control:

"Should Birth Control Be Legalized? (Women's Hour. January 31, 1946)

Discussed the pros and cons on birth control, its effect on preservation of mother and child health; the necessity of spacing children in order to provide better care; reasons for and against legalization of abortions".

As mentioned in Section 2.1, the wartime government implemented pro-natal policies and discouraged birth control. The above example illustrates the significant shift in public discourse around birth control since then. It demonstrates the social acceptance of birth control for the sake of maternal and child health.¹⁴ The next example challenges the patriarchal convention:

"The General Election and Man and Wife (Women's Hour. March 12, 1946)

While it may be praiseworthy for a wife to bow to her husband's will in many cases, the forthcoming general election demands that she make her own decision, entirely independent of others"

The program demands that women make their own choices of whom to vote. This goes against the conventional patriarchal view that a woman obeys her father, her husband, and her son. The final example then celebrates the improved status of women after World War II:

"Working Woman's Hour, September 10, 1950.

Equal rights for women in their working conditions was the subject of the principal feature on this program, in which Mrs. Hani

¹⁰ See Table E.10 on annual radio subscription fee from 1925 to 1955. Data are drawn from Okabe (2018).

¹¹ The head of the production team for women's programs, Fuji Egami recalls "All women's programs, including introductory announcements, were to be submitted to the Civil Information and Education Section 10 days before the broadcast. Translators kept typing all the time. All dramas, stories, lectures, interviews, debates and even round tables were stenographed, rewritten into Japanese, then translated into English. It took more than two weeks to broadcast programs after they were recorded. It was impossible to deliver timely information" (Egami, 1955), translated in English by the author.)

¹² See Figure E.4 NHK Yearbook (Japan Broadcasting Corporation, 1947, 1949) and GHQ/SCAP CIE Weekly Report (Radio Education Branch, 1946–1950).

¹³ Weekly Radio Report GHQ/SCAP CIE Box No. 5318 Folder 9.

¹⁴ General Headquarters (GHQ) explicitly stated they would not interfere with Japan's population problems or promote birth control. Yet they did not prohibit discussions regarding birth control — especially from perspectives of women's liberation, maternal protection, and children's health (Yamamoto, 2017). This radio example further illustrates this point.

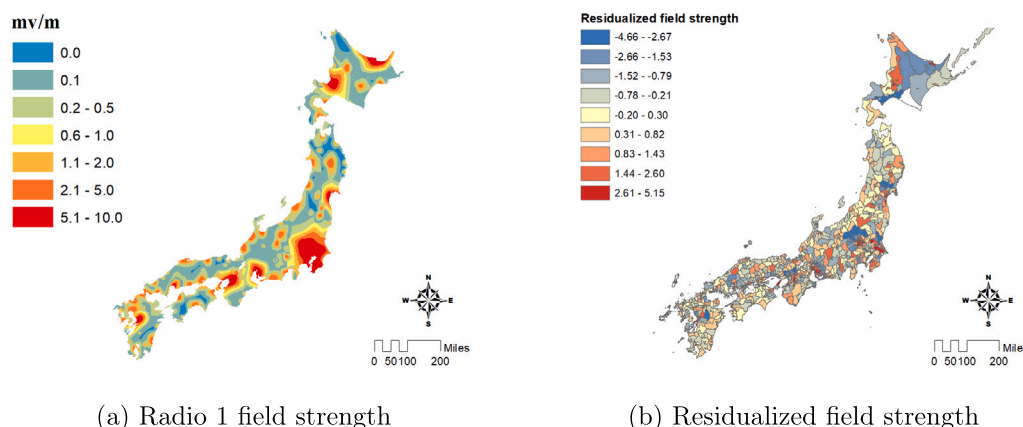


Fig. 1. Field strength and residualized field strength

Notes: Panel (a) shows the AM wave field strength, which serves as the instrumental variable for radio exposure. I digitized the map that the JBC published in 1949. Panel (b) shows the spatial distribution of residualized field strength. The residualized field strength is obtained by regressing field strength on the distance to a nearby transmitter, transmitter fixed effects, and all the other district characteristics variables such as population density and industrial composition. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

advised working women to feel proud of their position among men where they are winning economic independence for the first time in Japanese history”.

To understand the topic composition of the women’s radio programs in a more systematic manner, I classify the daily contents of women’s programs using Latent Dirichlet allocation, and show the year-by-year topic composition (Figure E.2).¹⁵ I find that in 1946, women’s programs primarily focused on politics and elections, which is consistent with historians’ account (Okahara, 2007). Over the years, the content of these programs became more diverse, covering topics such as women’s organizations, the interests of young women and girls, child development, new labor and welfare laws, and information on food and health. Ultimately, the programs addressed three key aspects of women’s empowerment — political, economic, and bodily autonomy.

3. An economic perspective on women’s radio programs

Through the lens of Akerlof and Kranton (2000)’s economics of identity, the radio messages can be viewed as a tool for reshaping and coordinating women’s sense of self, namely gender identity norm, on a large scale. Women, previously seen as unfit for political participation, were now encouraged to be important political contributors; Discussions on gender equality in the workplace began to surface; Birth control, discouraged during wartime, was promoted as beneficial for maternal and child health. Through radio, listeners learned that traditional norms were being challenged and new ones promoted. The medium created a shared experience in which women became aware that others were hearing the same messages. This collective awareness may have reduced the psychological costs of engaging in behaviors that had previously been seen as socially unacceptable. As a result, women exposed to these messages may have acted differently than they otherwise would have.

Given that women’s radio programs covered three key aspects of women’s empowerment, political, economic, and bodily autonomy, I hypothesize the following: First, women’s electoral turnout increased with greater exposure to radio programs that urged women to vote in the 1946 general election. Second, women’s labor market participation rose in response to increased exposure to radio content discussing women’s careers and labor laws protecting women’s workplace rights. Third, the annual birth rate decreased with more exposure to programs

promoting the benefits of birth control and spacing for women’s and children’s health. However, it remains an empirical question whether these radio programs had the intended consequences. Internalized identity norms can be resistant to change, and radio campaigning might also backfire.

4. Data

I compile district-level data (*shi* and *gun*) on radio listenership, radio signal coverage, outcome variables, and other district characteristics from official election reports, population censuses, vital statistics, local newspapers, and publications of the Japan Broadcasting Corporation (JBC). The unit of observation is the 1950 district boundary, and data from other years are harmonized to this boundary, accounting for changes over time.¹⁶ All variables except electoral turnout cover the entire nation.

The degree of exposure to women’s radio programs is measured using the 1946 radio subscription rate, defined as the share of households with a radio subscription, based on data from the 1946 JBC yearbook. These figures come from Japan’s radio receiver licensing system, which recorded district-level subscription rates. This measure is essential for scaling in the analysis, as subscription rates in 1946 ranged from about 5 percent to nearly 80 percent, with a national average of 37.3 percent.

I construct an instrumental variable for radio exposure using AM wave field strength, digitized from a map published by the JBC in 1949 (Fig. 1(a)). According to Japan Broadcasting Corporation (1947), the measurements were taken in 1948, making this the first postwar field strength map. I calculate average field strength by district and measure the distance to the nearest transmitter using latitude and longitude from the 1947 JBC yearbook¹⁷ and shapefiles of administrative boundaries.¹⁸

I hand-collect turnout by sex for the April 10, 1946 House of Representatives election, which was the first after women’s suffrage, from local editions of three national newspapers and multiple prefectural newspapers published between April 12 and April 18, 1946.¹⁹ The

¹⁶ Small islands and districts with complex boundary changes are excluded from the sample.

¹⁷ Only two transmitters were added between 1946 and 1948 (from 111 to 113), so changes in transmitter location are unlikely to confound results.

¹⁸ Shapefiles are from Maruyama Lab, Tsukuba University, Japan.

¹⁹ Sources include the Yomiuri and its local editions in Iwate, Miyagi, Fukushima, Ibaraki, Gunma, Saitama, Chiba, and Nagano; the Asahi (April 11) and its local editions in Shizuoka, Shimane (April 18), Nagasaki, Oita,

¹⁵ For a detailed explanation, see Appendix E.2.

Table 1
Balance test.

Panel A							
	Political variables pre WWII				Political variables during WWII		
	(1) Turnout (1928)	(2) WSL district (1928)	(3) WSL vote share (1928)	(4) Turnout (1937)	(5) Turnout (1942)	(6) IRAA vote share (1942)	
Residuals	0.0138 (0.0138)	−0.0212 (0.0373)	−0.0537 (0.0782)	0.0173 (0.0210)	0.00537 (0.00890)	0.00879 (0.0162)	
Mean Dep. Var. Observations	0.82 662	0.09 662	0.10 60	0.78 638	0.85 662	0.68 662	
Panel B							
	Demographic and economic variables pre/during WWII				Pre US occupation outcomes		
	(1) Male to female ratio at birth (1935)	(2) Male to female ratio (1940)	(3) Gender industrial segregation (1940)	(4) Primary sector labor share (1940)	(5) Female employment rate (1940)	(6) Marriage rate (1935)	(7) Birth rate (1935)
Residuals	−0.580 (0.479)	−0.654 (0.412)	−0.00144 (0.00486)	−0.0152 (0.0621)	−0.00328 (0.0166)	0.159 (0.246)	0.233 (0.987)
Mean Dep. Var. Observations	105.25 662	97.95 658	0.08 658	0.55 658	0.38 658	8.56 662	32.71 662

Note: WSL – Women's Suffrage League; IRRA – Imperial Rule Assistance Association. This table presents regression results examining the relationship between pre-war district characteristics and the residual radio signal. The residual signal is constructed by regressing radio field strength on decile-binned distance to the nearest transmitter, radio station fixed effects, prefecture fixed effects, the logarithm of the number of households, and an indicator for severe WWII damage from air raids or the atomic bomb. This residual captures variation in radio signal strength attributable to soil conditions between the transmitter and the receiving district. The unit of observation is a district (*shi* or *gun*), using 1950 boundaries. In Panel A, Column 3 restricts the sample to districts with at least one WSL-endorsed candidate. Column 4 restricts the sample to districts where voting occurred, excluding districts where the number of candidates equaled the number of seats. All regressions are weighted by district population, and standard errors are clustered at the radio station level. See Appendix B for variable definitions and data sources.

dataset covers 662 districts in 26 prefectures, representing 54 percent of eligible voters.²⁰

Table A.2 reports summary statistics for the district-level radio subscription rate, residualized field strength (which provides as-good-as-random variation in radio signal due to soil conditions), and other radio-related characteristics for all districts (Column 1), districts in the election sample (Column 2), and districts outside the election sample (Column 3). All variables are statistically similar between in-sample and out-of-sample districts, which suggests no sample selection based on radio-related characteristics. In the later analyses, I focus on within-prefecture, across-district variation rather than variation across prefectures.

Other variables, including electoral turnout, election results, labor market participation, marriage, and fertility outcomes, are drawn from population censuses, vital statistics, and official election reports. In Table 1 Panel B Columns 5–7, I report prewar mean outcomes: the female employment share is 0.38 (excluding family employment), the marriage rate is 9 per 1000 population, and the birth rate is 33 per 1000 population. Table A.1 lists all data sources.

5. Empirical strategy

To examine the impacts of women's exposure to women's radio programs, I use a linear causal model relating exposure to the outcomes of interest at the district level. For a district j in period t ,

$$y_{j,t} = \beta_0 + \beta_1 r_j + \gamma x'_{j,t} + u_{j,t} \quad (1)$$

and Miyazaki; Yamagata Shinbun; Kitaguni Shibun; Yamanashi Nichi Nichi Shinbun (April 13); Chubu Nihon Shinbun; Ise Shinbun; Kyoto Shinbun; Bocho Shinbun (April 16); Tokushima Shinbun; Ehime Shinbun; and Saga Shinbun. Unless otherwise noted, the publication date is April 12, 1946. Shinbun means “newspaper” in Japanese.

²⁰ In 1946, all men and women aged 20 or above were eligible to vote and were automatically registered.

where $y_{j,t}$ is an outcome of interest, r_j is the degree of exposure to women's radio programs, and $x_{j,t}$ is a vector of district characteristics, such as the distance from the nearest radio station, radio station fixed effects, prefecture fixed effects, population size, and to the extent of damage (casualties) caused during World War II.

As explained in Section 4, I measure the key explanatory variable, women's exposure to women's radio programs (r_j , expressed in standard deviation units), using the district-level radio subscription rate, which is defined as the share of households with a radio subscription. Potential measurement concerns are addressed in Section 7.2. The main outcome variables are female voter turnout and the vote share of female candidates in the 1946 election, the birth rate in 1947 and 1950, and the female employment rate, marriage rate, and divorce rate in 1950. The coefficient on the degree of radio exposure, β_1 , is the parameter of interest and captures the effect of exposure to women's radio programs on each outcome.

The key empirical challenge to identify the causal effect β_1 is that radio exposure might be endogenous to women's outcomes through unobserved district characteristics. It would be possible that radio subscriptions are correlated with subscribers' unobserved characteristics, such as attitude toward female empowerment and/or the Allied Occupation, openness to new ideas, or local culture. For example, women with a greater interest in politics may subscribe to radio to obtain information on politics. Such a positive correlation between subscription and unobserved characteristics bias the causal effect β_1 .

To address the potential endogeneity issue, I instrument the exposure by quasi-random variation in radio signal quality during daytime hours, which is as good as random to potential subscribers but increases their likelihood of radio subscription. The metric of the radio reception I use is the ground wave field strength (hereafter field strength), which depends on the horizontal distance from a nearby transmitter, output power of the transmitter, wavelength, and ground conductivity between the transmitter and receiver. The ground conductivity measures how fast the AM radio wave can propagate through a given soil type and depends on the moisture and salt contents of the soil. The key idea is as follows. On the one hand, the distance to a nearby

transmitter, output power, or wavelength may be based on strategic considerations,²¹ and the ground conductivity is as good as random to potential subscribers. Therefore, after controlling for distance and transmitter fixed effects, the local variation in the field strength can serve as an instrumental variable for the radio subscription rate. My identification strategy is similar to Strömberg (2004) and Adena et al. (2015) which also exploits features of AM radio wave propagation. Thus, I run the first stage equation as follows:

$$r_j = \pi_0 + \pi_1 z_j + f(d_{j,c(j)}) + \delta_{c(j)} + \phi_{p(j)} + x'_j \eta + v_j \quad (2)$$

where z_j denotes field strength, $d_{j,c(j)}$ is the distance from district j to its nearest transmitter, $\delta_{c(j)}$ captures fixed effects of that transmitter, $\phi_{p(j)}$ is prefecture fixed effects,²² and $\delta_{c(j)}$ includes the log of the number of households in district j , as well as an indicator equal to one if the district was severely damaged by air raids or the atomic bomb during World War II. I control for distance nonlinearly by including dummy variables for each distance decile.

Fig. 1(b) shows the identifying variation, namely the residualized field strength after controlling for distance, transmitter fixed effects, and all other control variables. Areas with high residualized field strength, shown in dark blue, and those with low values, shown in dark red, are located next to each other. This visual pattern suggests no systematic spatial structure, supporting the assumption that the residualized field strength is as good as randomly assigned. As shown in Fig. 2, higher residualized field strength is associated with higher radio subscription rates, demonstrating the relevance of the instrument. More detailed validation is provided in the next section.

With the instrumental variable in hand, I estimate the causal impact of radio exposure on each outcome using two-stage least squares (TSLS) with robust standard errors. The second-stage regression is specified as

$$y_{j,t} = \beta_0 + \beta_1 \hat{r}_j + g(d_{j,c(j)}) + \delta_{c(j)} + \phi_{p(j)} + x'_j \kappa + u_{j,t} \quad (3)$$

where $\hat{r}_j$ is the predicted radio exposure from the first stage regression (2). The term $g(d_{j,c(j)})$ controls for distance to the nearest transmitter by including dummy variables for each distance decile. $\delta_{c(j)}$ are nearest-transmitter fixed effects, $\phi_{p(j)}$ are prefecture fixed effects, and x includes population size and the extent of wartime damage (measured by casualties). Standard errors are clustered at the transmitter level. The coefficient β_1 is the parameter of interest and measures the effect of a one standard deviation change in radio exposure, as predicted by residualized field strength, on the outcomes.

6. Findings

Instrument validity. The IV strategy in Section 5 requires that the instrument (the residual variation in the field strength) is uncorrelated with unobserved determinants of the outcome (exogeneity) and influences the outcome only through increased exposure to radio programs (excludability). A key concern is that radio transmitters may have been placed in districts whose characteristics also predict the outcomes of interest. For example, these districts may have been more politically active, held different gender attitudes, or had varying views toward the military government and, by extension, the U.S. occupation. However,

²¹ For example, radio transmitters may have been strategically placed in the area with higher political aspirations, higher aspiration for freedom of marriage, higher demand for birth control, higher potential supply of female labor force, and so forth. Although such a concern may be unwarranted given the historical background of radio, as described in Section 2, these unobserved characteristics of women may indirectly relate to transmitter locations through urbanness, and therefore it is still important to control for the distance.

²² In Japan, a prefecture is the primary administrative division, governed by an elected governor, legislature, and supporting bureaucracy. Districts are smaller geographical units within prefectures.

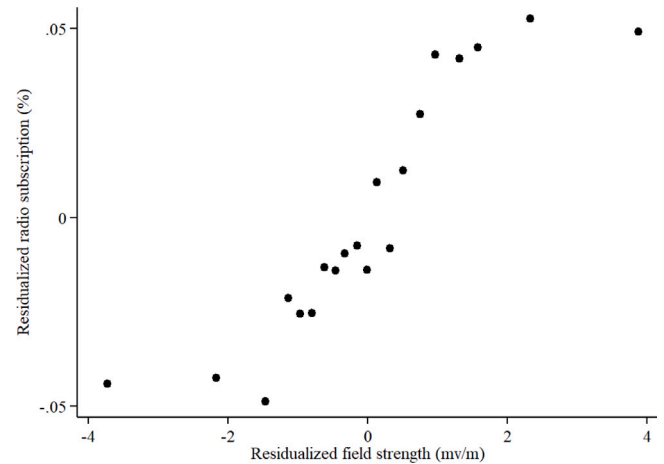


Fig. 2. Visualization of the first stage

Notes: This figure shows a binscatter plot of the residualized radio subscription rate against the field strength with a minimum set of control variables, i.e., the distance to a nearby transmitter and transmitter fixed effects. Each bin represents about 40 districts. This figure is a graphical depiction of the first stage: as we expect, greater residualized field strength does increase radio subscriptions. The S-shaped curve implies that the extremely low [high] field strength does not affect radio subscriptions because people there cannot [can] listen to radio anyway.

this type of strategic placement is less likely, as transmitters were not installed by the U.S. occupying authority, as explained in Section 2.2.

To further assess validity, I test whether pre-U.S. occupation district characteristics are correlated with the residualized radio signal. I focus on four potential confounding pathways: (i) gender attitudes and women's empowerment, (ii) democratic practices, (iii) support for the wartime militaristic government, and (iv) economic development. Below I summarize the approach; details on data sources and variable construction are in Appendix B.

First, districts with stronger signals might differ in gender attitudes and women's empowerment. To proxy this, I construct several indicators: (a) whether a male candidate publicly supported women's suffrage in the 1929 election and his vote share; (b) the sex ratio in reported births from the 1935 vital statistics, reflecting son preference; and (c) an industrial sex segregation index measuring the extent of gender overlap in non-home employment. Second, districts with stronger signals may have had more established democratic practices, potentially shaping turnout through habit formation. I proxy this with district-level turnout in the 1928 general election (the first after male universal suffrage) and the 1937 election (the last before World War II). Third, districts with stronger signals may have varied in their support for the wartime militaristic government, shaping attitudes toward the U.S. occupation. I proxy this with turnout and the vote share of the Imperial Rule Assistance Association in the 1942 wartime election, when existing parties were replaced to mobilize war support. Fourth, districts with stronger signals may have been at different stages of economic development, particularly in their reliance on primary industries. I proxy this with the labor share in agriculture, fishing, and mining from the 1940 Population Census.

Table 1 shows that none of these variables are significantly correlated with residualized field strength in pairwise comparisons. In a joint regression, the covariates explain only 3 percent of the variation ($R^2 = 0.03$), and no coefficients are significant (Column 3 of Table B.3). Repeating the exercise on the electoral subsample yields the same conclusion: no significant correlations with residualized signal strength (Column 1). Taken together, these results provide support for the validity of the instrument.

Table 2
Impacts of women's radio: Voter participation.

	Radio subscription in std.dev. unit	Women's turnout			Men's turnout		
	(1) FS	(2) RF	(3) OLS	(4) TSLS	(5) RF	(6) OLS	(7) TSLS
Field strength in std.dev. unit	0.571*** (0.105)	0.0139* (0.00587)			0.00514 (0.00520)		
Radio subscription in std.dev. unit			0.0257*** (0.00555)	0.0244** (0.00839)		−0.000272 (0.00451)	0.00934 (0.00804)
Observations	337	337	337	337	327	327	327
First stage F-stat				29.33			28.16
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome		0.67	0.67	0.67	0.81	0.81	0.81

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: FS = first stage; RF = reduced form; OLS = ordinary least squares; TSLS = two-stage least squares. This table reports first stage, reduced form, OLS, and TSLS estimates of the effect of radio exposure on women's turnout (Columns 2–4) and men's turnout (Columns 5–7) in the 1946 general election. Column 1 shows the first-stage effects on radio subscription rates. The sample for men's turnout is smaller because in one prefecture (Miyazaki), the local newspaper reported turnout rates only for women, not men. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the nearest radio station level.

First stage. Fig. 2 shows a binscatter plot of the residualized radio subscription rate against the residualized field strength, showing a graphical representation of the first stage of my IV model. Column 1 of Table 2 reports the first stage estimate with the minimal set of covariates: after controlling for the transmitter fixed effect and distance to a nearby transmitter, a one standard deviation of field strength increases radio subscription by 0.57 standard deviation unit.

Impacts on voter participation. I first present the results of voter participation in 1946 to illuminate the immediate impact of radio exposure. I find that exposure to women's radio programs yields positive impacts on women's political participation. Table 2 Column 4 shows a two-stage least-square estimate of the coefficient on the radio exposure in regression Eq. (1). A one standard deviation increase in radio subscriptions increases women's electoral turnout by 2.4 percentage points.

Meanwhile, Column 7 presents the TSLS estimate regressing men's turnout on the radio subscription. Similar to the OLS estimate, the TSLS estimate is not statistically distinguishable from zero at the five percent level. The positive gap between women's turnout (Column 5) and men's turnout (Column 7) suggests that the radio subscription has impacts only on women through the provision of women's radio programs.

The estimated impact is both statistically and substantively significant: it explains 31 percent of the standard deviation in women's turnout. The implied persuasion rate (DellaVigna and Kaplan, 2007), which is the share of eligible voters who changed their behavior due to radio exposure among those at risk of changing, is 18.1%.²³ For comparison, prior studies estimate persuasion rates of mass media on turnout in historical contexts between 4.4% for television (Gentzkow, 2006) and 12.8% for local newspapers (Gentzkow et al., 2011). The effect I find is larger than these historical benchmarks and closer in magnitude to the impacts typically achieved by effective Get-Out-the-Vote (GOTV) campaigns: 15.6% for door-to-door canvassing (Gerber and Green, 2000) and 20.4% for phone calls by youth volunteers (Green and Gerber, 2001) (see DellaVigna and Gentzkow (2010) for review). One likely reason for the relatively large effect in my context is that newly enfranchised women lacked prior voting histories, making their turnout decisions especially malleable.

²³ The share of eligible voters who turned out because of radio exposure is 0.07, and the share of voters who were at risk of changing their behavior is 0.41.

Impacts on vote share. Table 3 reports the impacts of increased radio exposure on vote shares of female candidates. As Column 3 shows, in areas with greater radio exposure by a one standard deviation received more vote by 2.3 percentage points. The estimate implies a persuasion rate of 4.6%.²⁴ This falls within the range documented in previous studies, which vary from 2.0% for the unsurprising Democratic endorsement on support for Gore to 11.6% for the availability of Fox News on Republican vote share (see DellaVigna and Gentzkow (2010) for review).

One can calculate the counterfactual election outcome with zero radio exposure in every district. Women's share among winners would have been only 5.3 percent, compared to the actual representation of 8.2 percent. This finding makes a case against the *double-vote hypothesis*, which posited that women, especially married women, would simply cast the same votes as their husbands and have no influence on electoral outcomes. My finding, by contrast, highlights that women's votes matter.

Employment and family formation. Turning to the effects on women's labor force participation, Upper Panel Column 2 of Table 4 shows that the estimated impact is statistically indistinguishable from zero.

Finally, I turn to the impact of women's programs on family formation. As Table 4 shows, both impacts on the marriage rate (Upper Panel Column 5) and divorce rate (Column 8) are near zero and statistically indistinguishable from zero.

Radio exposure does have impacts on birth rate, which gradually emerges. In 1947, the estimated impact of a one unit standard deviation radio exposure is −0.7 births per 1000 population, albeit statistically significant. In 1950, five years after the radio programming started, a one unit standard deviation radio exposure affects the crude birth rate by 2.0 (Table 4 Lower Panel Column 5), which is statistically significant at one percent.

Additionally, I show that the decline in live births is not due to the change in stillbirths (fetal deaths). One may worry that radio exposure may change stillbirths while the total births remain constant.

²⁴ Following DellaVigna and Kaplan (2007), I compute the persuasion rate for vote share as the proportion of votes cast due to radio exposure, relative to the share of votes at risk of being persuaded, adjusted for overall voter turnout.

Table 3
Impacts of women's radio: Female candidates' vote share.

	Female candidate's vote share			
	(1) RF	(2) OLS	(3) TSLs	(4) TSLs
Field strength in std.dev. unit	0.0143* (0.00684)			
Radio subscription in std.dev. unit		0.00488 (0.00323)	0.0230* (0.00986)	
Women's turnout %				0.00711 (0.00455)
Observations	618	618	618	618
First stage F-stat			19.10	9.75
Candidate FE	Yes	Yes	Yes	Yes
Prefecture FE	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes
Distance control	Decile bins	Decile bins	Decile bins	Decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes
Mean outcome	0.08	0.08	0.08	0.08

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: RF = reduced form; OLS = ordinary least squares; TSLs = two-stage least squares. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the candidate level.

For example, radio exposure may raise women's health awareness, which may prevent fetal deaths. Alternatively, radio exposure may inform women about the Eugenic Protection Law of 1948. Since the law legalized abortions, informed women may be more likely to access abortions. To best address such a concern, I examine the impact of radio exposure on stillbirths, which include both induced abortions and stillbirths due to natural causes after 12 weeks of pregnancy.²⁵ The lower panel of Table 4 Column 7 shows that the impact on the number of stillbirths per 1000 births is 0.1 and not statistically significant.²⁶ Thus, the decline in live births is not likely due to the significant change in stillbirths.

Summary. Women's radio programs contemporaneously amplified women's voices in the political sphere in Japan: the women's programs effectively lead more women to vote, which in turn resulted in closing the gender representation gap. My findings are also consistent with what the GHQ/SCAP Radio Unit wrote in a weekly radio report: the women's programs "undoubtedly contributed in a large measure to the fact that 65 percent of the eligible women voters went to the polls".²⁷

Radio exposure has also played a role in accelerating fertility decline. The results above suggest that the decline is due neither to an increase in women's labor force participation, to a decrease in marriages, or an increase in the utilization of induced abortions. Therefore, the fertility decline can be attributed to the remaining mechanism, that is, increased family planning. This decrease can be seen as beneficial for women as it reduces health risks and child-rearing burdens. Furthermore, it holds significant implications for society at large.

The null result on female employment could be due to a tightened labor market following the influx of men returning from World War II. It could also be due to the persistent traditional attitudes of male-dominated employers and men's view on working women remained unchanged. This is plausible, considering that the radio campaigns were primarily directed at women, possibly overlooking the need to simultaneously change men's norms.

²⁵ No data available on abortions and stillbirths due to natural causes separately.

²⁶ The number of births is the sum of the number of live births and stillbirths. Data on stillbirths are not available for the year 1947.

²⁷ Weekly radio report, SCAP Civil Information and Education Section.

7. Discussion

Overall, my findings indicate that greater exposure to women's radio programs influenced women's political outcomes and fertility during the Allied Occupation. In this section, I discuss potential mechanisms, examine possible longer-run effects beyond the Occupation period, and consider the generalizability of the results.

7.1. Robustness

To assess the robustness of my main findings, I re-estimate the main specification controlling for prewar outcomes, as well as for prewar outcomes together with additional district-level demographic characteristics. These characteristics capture prewar economic development and gendered outcomes, including the sex ratio at birth, industrial sex segregation, the overall sex ratio, and the primary-sector labor share. Results are reported in Tables C.4 and C.5, and remain robust overall.

I also re-estimate the main specification using spatial standard errors following Conley (1999). This approach allows for spatial correlation that decays with distance, using a weighting function defined by a kernel in each dimension of the Cartesian plane (north-south and east-west). The kernel equals one at the origin, decreases linearly to zero at a cutoff, and remains zero beyond that point. I measure economic distance by physical distance and set the cutoff at 30 km, approximately the average distance between neighboring districts. Results, reported in Table C.6, show that the main findings remain robust when accounting for spatial correlation.

7.2. Mechanisms

In Section 3, I have proposed that women's radio programs that broadcast empowerment-oriented messages could influence women's outcomes by reshaping gender identity norms on a large scale. In this subsection, I present correlational evidence linking radio exposure to gender norms, and consider two alternative interpretations by examining heterogeneous effects across districts with different male-to-female ratios, which varied significantly due to men's wartime participation and casualties.

Radio exposure and gender norms. In Appendix E.6, I present correlational evidence that frequent radio exposure is positively associated with gender-egalitarian views (approval of women working outside

Table 4

Impacts of Women's Radio: Employment, Marriage, Divorce, and Birth Rate.

	Women's employment rate exclud. family emp			No. marriages per 1000 population in 1950			No. divorces per 1000 population in 1950		
	(1) OLS	(2) TSLS	(3) RF	(4) OLS	(5) TSLS	(6) RF	(7) OLS	(8) TSLS	(9) RF
Radio subscription in std.dev. unit	-0.00168 (0.00348)	-0.00630 (0.0113)		-0.0899 (0.0944)	0.0289 (0.178)		-0.0106 (0.0273)	0.0158 (0.0348)	
Field strength in std.dev. unit			-0.00342 (0.00694)			0.0156 (0.108)			0.00852 (0.0216)
Observations	662	662	662	660	660	660	659	659	659
First stage F-stat		36.41			34.22			34.08	
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Distance control	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome	0.28	0.28	0.28	8.42	8.42	8.42	0.97	0.97	0.97
	No. live births per 1000 population in 1947			No. live births per 1000 population in 1950			No. still births per 1000 population in 1950		
	(1) OLS	(2) TSLS	(3) RF	(4) OLS	(5) TSLS	(6) RF	(7) TSLS		
Radio subscription in std.dev. unit	-0.355 (0.209)	-0.698 (0.450)		-1.249*** (0.341)	-2.034*** (0.514)		-0.0788 (0.115)		
Field strength in std.dev. unit			-0.379 (0.257)			-1.103** (0.333)			
Observations	662	662	662	662	662	662	661		
First stage F-stat		36.41			36.41		34.15		
Prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Transmitter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Distance control	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins	Decile bins		
Log N of hh and WWII damage	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
Mean outcome	34.54	34.54	34.54	28.14	28.14	28.14	2.64		

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

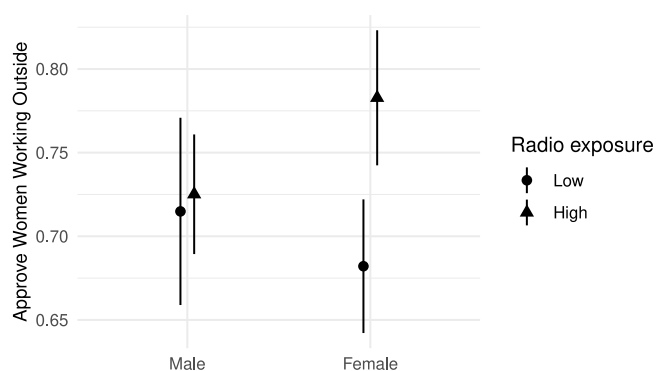
Note: OLS = Ordinary Least Squares; TSLS = Two-Stage Least Squares; RF = Reduced Form. All regressions include nearest transmitter fixed effects, prefecture fixed effects, distance to the nearest transmitter in deciles, and log population size as controls. Standard errors are clustered at the nearest radio station level.

the home) among women but not among men. Fig. 3 provides a visual summary of this pattern. The survey was conducted in 1953 by the Institute of Statistical Mathematics in Japan, just a year after the U.S. occupation ended. It is the earliest survey I could find that asked about both radio listenership and gender attitudes, with micro-level data available.²⁸ Although the evidence is correlational, the results indicate that women who reported frequent radio exposure were more likely to endorse egalitarian views, whereas men's views were unaffected. This pattern is consistent with the original hypothesis that radio exposure contributed to shifts in women's gender norms.

Women's program versus general program. Since radio exposure is measured using subscription rates, one alternative interpretation is that the estimated effects may capture exposure to general radio programming, such as daily news, rather than content specifically aimed at women.

However, this possibility appears unlikely. As shown in Table 2 Column 7, radio exposure has no significant impact on men's turnout. This serves as a useful test for the influence of general programming. Since women's programs were designed for female audiences and men listened to them far less frequently, if at all (see Figure E.3), the absence of an effect among men suggests that general radio content alone did not substantially mobilize voters.

²⁸ Because the survey lacks information on respondents' place of residence, I cannot use location-based variation in field strength as an instrument for radio listenership.

**Fig. 3.** Radio exposure and gender norms by sex

Notes: The figure shows the predicted probability of approving women working outside the home by gender (x-axis) and level of radio exposure (distinguished by marker shape). Each point corresponds to the model-predicted approval rate for the given subgroup, and the vertical bars show 95% confidence intervals around those predictions. The regression result is presented in Appendix E.6.

To extend this logic and conclude that women, too, were unaffected by general radio programming, we must assume that men and women were equally susceptible, or equally unresponsive, to general information transmitted via radio. This assumption could be challenged if newly enfranchised voters were systematically more responsive to

radio, since in 1946 all women were voting for the first time, whereas men over age 25 had been eligible to vote since the prewar period.

Yet, my data suggest this concern is not warranted in the context I study. In Appendix D, I examine whether radio exposure increased men's turnout in the 1928 election, which was the first general election held after all adult men gained voting rights. I find no effect, indicating that newly enfranchised men in 1928 were no more responsive to radio exposure than experienced voters. This finding reduces the concern that newly enfranchised voters are particularly malleable to radio influence, at least in the Japanese electoral context of the time.

Taken together, the evidence supports the interpretation that the primary impact of radio on women's turnout in 1946 was driven by the content of women-specific programming. This is perhaps unsurprising, given that women's programs, unlike general broadcasts, covered topics highly relevant to women, such as profiles of female candidates, labor laws affecting women, the right to choose marriage partners, and the health benefits of birth spacing.

Normative message versus informational content. While the women's radio programs intended to dismantle the existing patriarchal norms, there is also a possibility that radio may have provided new information to women. The informational channel would be particularly probable if the radio campaign affected more illiterate and less-educated women, who were more informationally constrained. To assess whether the radio impacts differ based on the district's level of educated women, I estimate the following regression:

$$y_{j,1946} = \beta_0 + \beta_1 r_{j,1946} + \beta_2 r_{j,1946} \times z_j + \beta_3 z_j + \gamma x'_{j,1946} + u_{j,1946} \quad (4)$$

where z_j is an indicator equal to one if the share of highly educated women (10 or more years of education) is above the median. I instrument radio exposure with field strength. As shown in Figure C.1, the average residual field strength does not differ, the average residual field strength does not differ between districts with above-median and below-median shares of highly educated women.

The third rows of Table C.7, Column 2, and Table C.8, Column 2, present the heterogeneous effects of women's radio programming on turnout and fertility, respectively, by the share of highly educated women in a district. The effect of radio exposure on turnout does not vary with women's educational attainment. For fertility, a higher share of highly educated women slightly amplifies the radio effect, but not vice versa. These results do not support the hypothesis that radio primarily served to inform illiterate or less-educated women.

Similarly, I test whether the impacts differ between urban and rural districts by setting z_j in Eq. (4) to an indicator equal to one if district j is a city. This heterogeneity analysis examines whether radio exposure might have had a larger effect in rural areas ($z_j = 0$), where access to information may have been more limited. However, I find no statistically significant heterogeneous effects (see the 5th row of Table C.7, Column 2 for turnout, and Table C.8, Column 2 for fertility). Again, this result provides little support for the view that the informational content of women's radio programming is the primary driver of my main findings, and instead favors the original hypothesis that normative messages aimed at dismantling pre-war patriarchal norms plays a key role.

Male to female ratio. Finally, I explore the possibility of heterogeneous impact based on the male to female ratio. Due to military casualties, the male to female ratio varies substantially with the mean of 0.92 and the standard deviation of 0.06. Asai and Kambayashi (2023) document that such across-district variation comes from *home regiments*: each regiment consisted of men from the same prefecture and it was as if home away from home. Moreover by chance, some regiments were heavily damaged. This is because the allies employed the leapfrogging strategy, i.e., bypassing heavily fortified islands instead of capturing every island in sequence. Therefore, variation in the male-to-female ratio in the aftermath of WWII is as-good-as-random, and provides the opportunity to test the heterogeneous effect of radio exposure.

Different theories predict heterogeneous impacts in opposite directions. On the one hand, marriage bargaining theory suggests that female scarcity should yield a positive coefficient β_2 . The reasoning is that when women are scarce, their bargaining power increases, which could amplify the effect of radio exposure. Whether this mechanism operates in the present setting is ultimately an empirical question. On the other hand, prior studies such as Grosjean and Khattar (2019) show that male-biased sex ratios tend to sustain conservative gender norms. If this mechanism dominates, male abundance would suppress the effects of radio in areas with higher male-to-female ratios.

The seventh rows of Table C.7, Column 4, and Table C.8, Column 4, report the heterogeneous effects of women's radio programming on turnout and fertility, respectively, by the male-to-female ratio. A one standard deviation increase in the male-to-female ratio reduces the impact of radio exposure by 2.7 percentage points. In other words, radio exposure has a smaller effect on women's turnout when men are more abundant and women are relatively scarce. For fertility, the point estimate is in the same direction, although not statistically significant. Overall, these findings are more consistent with the mechanism that male abundance sustains conservative gender norms than with the marriage bargaining theory.

7.3. Possibilities of long-run consequences

Beyond the immediate impacts, it is relevant to ask how the effects evolved in the medium and long run. One question is whether the mid-run effects of the radio programs were sustained, faded, or amplified. Using qualitative evidence and aggregate quantitative data, I suggest that the impacts were complex. Appendix E.4 shows that women's turnout increased in the 1949 and 1952 House of Representatives elections, with low-exposure areas catching up to high-exposure areas, likely due to spillovers and social sanctions. At the same time, qualitative accounts suggest resistance to women's growing political presence. MacArthur reportedly "warned them [female parliamentarians] from forming a female bloc" (Ichikawa, 1977; Ogawa, 1997), and an electoral reform in 1947 was expected to disadvantage women's representation.²⁹

Long-run and intergenerational effects also represent an avenue for future research beyond the scope of this study. Women who listened to the radio programs may have raised their daughters with different expectations, and sons could likewise have served as an important channel. In the United States, Fernández et al. (2004) show that the wives of men whose mothers worked during World War II were significantly more likely to work, suggesting that norms can be transmitted through family dynamics and shape the next generation of women. A similar mechanism may have operated in my context. Examining such effects, however, would require contemporary individual-level data with detailed childhood geographic information, which are currently unavailable. Access to such data could shed light on radio's influence on marriage timing, partner matching, fertility, birth spacing, and regional migration patterns. Japan's rapid post-Occupation economic growth also presents an interesting case for studying the relationship between women's empowerment and economic growth, where the evidence remains mixed and further research is warranted (Duflo, 2012; Anderson, 2022).

7.4. Generalizability

Finally, my findings offer broader implications in two directions, though with important caveats. First, they speak to the potential of

²⁹ For example, during the March 28, 1947 committee meeting, Kato Chuzo of the Social Democratic Party of Japan warned that the reform would reduce female representatives, and Ono Banboku of the Japan Liberal Party reportedly told women parliamentarians it might be their last chance to serve (The House of Representatives, 1947a). Hara Hyonosuke raised the same concern on March 30, 1947 (The House of Representatives, 1947b).

mass media to empower women and other underserved populations. While radio is a legacy medium, it remains widely used and influential in many parts of the world today. Thus, the core insights of this study would remain relevant.

That said, the context of Occupied Japan is unique as is always the case in an empirical study. Female empowerment is multidimensional (Anderson, 2022), and the radio programming I study delivered bundled messages spanning education, work, and civic life. This message bundling may have contributed to the observed impacts. Furthermore, postwar Japan was undergoing a profound societal rethinking, with citizens reevaluating prewar norms. Such a moment of ideological openness may have made listeners particularly receptive to new messages, enhancing the efficacy of the broadcasts.

It is also important to note that the women's radio programs I study targeted women exclusively. To achieve greater and more sustained impacts, it may be necessary to engage men and the broader environment surrounding women. If pro-gender-equality messages had been directed at both women and men, they might have reinforced each other, although they could also have provoked backlash. One-sided messaging may also have contributed to diverging gender attitudes between men and women, potentially generating conflict. Women's radio programs were widespread internationally,³⁰ but their design and audiences varied. An instructive comparison comes from Sweden, a country now known for its gender egalitarianism, where in the 1960s campaigns promoted married women's employment not only through radio but also through evening television programs that families could watch together. These broadcasts explicitly linked women's work outside the home to changes in family organization (Lundqvist, 2017). This cross-country comparison may point to the importance of considering both content and audience.

Second, the study offers insights into the role of foreign-sponsored mass media in post-conflict settings and suggests that informational outreach to populations marginalized under the previous regime can have important implications for nation rebuilding. The radio programs examined here challenged gender disparities, but similar platforms have also been used to promote a wide range of values and information. For example, the United States has operated Voice of America, its largest and longest-running international broadcaster, since 1942, and the BBC has long served as a global source of cross-border news. These precedents underscore the potential of foreign-sponsored media, as in my case, to influence social norms during critical periods of political and social reconstruction.

At the same time, my findings should not be interpreted as a blanket endorsement of foreign media intervention, nor as an argument for gender empowerment through military occupation. Rather, this study aims to understand how media exposure, under specific historical circumstances, affected women's outcomes.

Japan's case underscores the importance of local agency in shaping outcomes. As discussed in Section 2, women's programs were produced through collaboration between American and Japanese women. Although final approval rested with the occupying authorities, historians note that Japanese women played a central role in shaping and localizing the content, at times resisting imposed narratives. American contributors also made notable efforts to understand Japanese society and culture. The result was far from a unilateral imposition of norms. These production dynamics are crucial for assessing the functioning of foreign media policies, and future research should investigate the conditions under which such collaborations produce meaningful, locally resonant change.

8. Conclusion

This study examines the impact of women's radio programs broadcast by the US-led occupying force in Japan between 1945 and 1952 to dismantle prewar patriarchal norms. Using local variation in radio signal strength driven by soil conditions as an instrumental variable, I provide causal evidence that greater exposure to these programs increased women's electoral turnout. Radio exposure also accelerated fertility decline, not through changes in marriage rates, or greater use of induced abortions, but through the adoption of family planning. This decline benefited women by reducing health risks and child-rearing burdens and carried broader social implications. By contrast, I find no effect on female employment, which may reflect the tightened labor market following the return of men from World War II or the persistence of traditional attitudes among male-dominated employers. The latter possibility suggests that, because the radio campaigns were directed primarily at women, they may have overlooked the need to shift men's norms as well.

Taken together, my findings suggest that pro-gender-equality messages directed at women have empowered Japanese women in the aftermath of World War II. This not only has significant implications for women's lives, but also impacts society by shifting the political landscape and accelerating fertility transition, potentially paving the way for the rapid economic growth that followed. Thus the empowerment of women concerns not just women but society at large.

My findings open new avenues of economic research. First, the case of Occupied Japan limits my ability to investigate what would have happened if both men and women, or only men, had been exposed to women's radio content. This limitation, however, provides motivation for field experiments to understand the nature of targeted mass media interventions further. Second, women who are exposed to new gender norms may have passed them onto their children even if they themselves did not change their behavior. Such an inter-generational transmission of the impact of norm-based interventions would be an interesting avenue of future research.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT 4o in order to check grammar and spelling. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jdevco.2025.103620>.

Data availability

Data will be made available on request.

³⁰ Examples include Woman's Hour on BBC Radio 4 in the UK (from 1946), Radio Donna in Italy (from 1976), Womankind in the US (1969), Hemmafru byter yrke in Sweden (1965), and RadiOrakel in Norway (from 1982).

References

- Adena, Maja, Enikolopov, Ruben, Petrova, Maria, Santarosa, Veronica, Zhuravskaya, Ekaterina, 2015. Radio and the rise of the nazis in prewar Germany. *Q. J. Econ.* 1885–1939.
- Akerlof, George A., Kranton, Rachel E., 2000. Economics and identity. *Q. J. Econ.*
- Anderson, Siwan, 2022. Unbundling female empowerment. *Can. J. Econ./Revue Can. D'Écon.* 55 (4), 1671–1701.
- Asai, Kentaro, Kambayashi, Ryo, 2023. The Consequences of Hometown Regiment: What Happened in Hometown When the Soldiers Never Returned? Technical Report, Institute of Economic Research, Hitotsubashi University.
- Bailey, Martha J., 2006. More power to the pill: The impact of contraceptive freedom on women's life cycle labor supply. *Q. J. Econ.* 121 (1), 289–320.
- Beach, Brian, Hanlon, W. Walker, 2022. Culture and the historical fertility transition. *Rev. Econ. Stud.* 90 (4), 1669–1700.
- Blouin, Arthur, Mukand, Sharun W., 2019. Erasing ethnicity? Propaganda, nation building, and identity in rwanda. *J. Political Econ.* 127 (3), 1008–1062.
- Cheung, Maria, 2012. Edutainment radio, women's status and primary school participation: Evidence from Cambodia.
- Conley, T.G., 1999. GMM estimation with cross sectional dependence. *J. Econometrics* 92 (1), 1–45.
- Corder, J. Kevin, Wolbrecht, Christina, 2016. Counting Women's Ballots: Female Voters from Suffrage through the New Deal. In: *Cambridge Studies in Gender and Politics*, Cambridge University Press.
- DellaVigna, Stefano, Enikolopov, Ruben, Mironova, Vera, Petrova, Maria, Zhuravskaya, Ekaterina, 2014. Cross-border media and nationalism: Evidence from serbian radio in croatia. *Am. Econ. J.: Appl. Econ.* 6 (3), 103–132.
- DellaVigna, Stefano, Gentzkow, Matthew, 2010. Persuasion: Empirical evidence. *Annu. Rev. Econ.*
- DellaVigna, Stefano, Kaplan, Ethan, 2007. The fox news effect: Media bias and voting. *Q. J. Econ.* 122 (3), 1187–1234.
- Doepke, Matthias, Kindermann, Fabian, 2019. Bargaining over babies: Theory, evidence, and policy implications. *Am. Econ. Rev.* 109 (9), 3264–3306.
- Duflo, Esther, 2012. Women empowerment and economic development. *J. Econ. Lit.* 50 (4), 1051–1079.
- Egami, Fuji, 1955. Fujin no jikan no ju nen [ten years of women's hour]. *Fujin Koron* p.164–165.
- Fernández, Raquel, Fogli, Alessandra, Olivetti, Claudia, 2004. Mothers and sons: Preference formation and female labor force dynamics. *Q. J. Econ.* 119 (4), 1249–1299, Publisher: Oxford University Press.
- Gagliarducci, Stefano, Onorato, Massimiliano Gaetano, Sobbrío, Francesco, Tabellini, Guido, 2020. War of the waves: Radio and resistance during world war II. *Am. Econ. J.: Appl. Econ.*
- Gentzkow, Matthew, 2006. Television and voter turnout. *Q. J. Econ.* 121 (3), 931–972.
- Gentzkow, Matthew, Shapiro, Jesse M., Sinkinson, Michael, 2011. The effect of newspaper entry and exit on electoral politics. *Am. Econ. Rev.* 101 (7), 2980–3018.
- Gerber, Alan S., Green, Donald P., 2000. The effects of canvassing, telephone calls, and direct mail on voter turnout: A field experiment. *Am. Political Sci. Rev.* 94 (3), 653–663.
- Giuliano, Paola, 2020. Gender and culture. *Oxf. Rev. Econ. Policy* 36 (4), 944–961.
- Goldin, Claudia, Katz, Lawrence F., 2002. The power of the pill: Oral contraceptives and women's career and marriage decisions. *J. Political Econ.* 110 (4), 730–770.
- Green, Donald P., Gerber, Alan S., 2001. Getting out the youth vote: Results from randomized field experiments. *Yale Univ.*
- Grosjean, Pauline, Khattar, Rose, 2019. It's raining men! Hallelujah? The long-run consequences of male-biased sex ratios. *Rev. Econ. Stud.* 86 (2), 723–754.
- Ichikawa, Fusae, 1977. Nihon fujin mondai shiryō shusei [collection of materials relating to women's issues in Japan].
- Japan Broadcasting Corporation, 1947. Radio Yearbook (Radio Nen Kan). Technical Report, Tokyo.
- Japan Broadcasting Corporation, 1949. Radio Yearbook (Radio Nen Kan). Technical Report, Tokyo.
- Japan Broadcasting Corporation, 1950. Radio Yearbook (Radio Nen Kan). Technical Report, Tokyo.
- Jensen, Robert, Oster, Emily, 2009. The Power of TV: Cable Television and Women's Status in India. Technical Report, vol. 3, pp. 1057–1094.
- Jiang, Hua, 2022. Koto Jogakko ni okeru Ryosai Kenbo Kyoiku no Seiritsu to Tenkai [The Establishment and Development of Good Wife and Wise Mother Education at Girl's High School]. Toshindo, Tokyo.
- Kearney, Melissa S., Levine, Phillip B., 2015. Media influences on social outcomes: The impact of MTV's 16 and pregnant on teen childbearing. *Am. Econ. Rev.* 105 (12), 3597–3632.
- Khalifa, Suzanna, 2022. Female genital cutting and bride price.
- Kobayashi, Yoshie, 2004. A Path Toward Gender Equality: State Feminism in Japan. Routledge.
- Koyama, Shizuko, 1991. Ryosai Kenbo toiu Kihan [The norm of a Good Wife and Wise Mother]. Keiso Shobo, Tokyo.
- La Ferrara, Eliana, Chong, Alberto, Duryea, Suzanne, 2012. Soap operas and fertility: Evidence from Brazil. *Am. Econ. J.: Appl. Econ.* 2012 (4), 1–31.
- Lowes, Sara, 2022. Culture in historical political economy. In: *Working Paper Series*, (30511).
- Lundqvist, Åsa, 2017. Activation through information and persuasion. In: *Transforming Gender and Family Relations*. Edward Elgar Publishing, pp. 112–138.
- Luther, Catherine A., Boyd, Douglas A., 1997. American occupation control over broadcasting in Japan, from 1945 to 1952. *J. Commun.* 47 (2), 39–59.
- Mayo, J. Marlene, 1988. The war of words continues: American radio guidance in occupied Japan. In: *The Occupation of Japan: Arts and Culture*. Norfolk, VA.
- Ogawa, Manako, 1997. The Establishment of the Women's and Minors' Bureau in Japan. *U.S.-Japan Women's J. Engl. Suppl.* 13 (13), 56–86.
- Ogino, Miho, 2008. Kazoku keikaku e no michi: kindai nihon no seishoku o meguru seiji [the road to "family planning": politics over reproduction in modern japan]. Iwanami Shoten.
- Okabe, Tadanobu, 2018. The 50 years of Japanese Radio 1925-75. Japan Radio Museum, Matsumoto, Japan.
- Okahara, Miyako, 2007. Amerika senryoki no minshuka seisaku: Rajio hoso ni yoru nihon jōsei saikyōiku puroguramu [democratization policies during the US occupation period: Radio broadcasting programs for the re-education of Japanese women].
- Olken, Benjamin A., 2009. Do television and radio destroy social capital? Evidence from indonesian villages. *Am. Econ. J.: Appl. Econ.* 1 (4), 1–33.
- Pharr, Susan, 1987. The politics of women's rights. In: *Policy and Planning During the Allied Occupation of Japan*. University Press of Hawaii, chapter 8.
- Russo, Gianluca, 2019. Mass media and cultural homogenization: Broadcasting the American dream on the radio.
- Silva, Tiloka De, Tenreiro, Silvana, 2017. Population Control Policies and Fertility Convergence. *J. Econ. Perspect.*
- Smulyan, Susan, 2002. Now it can be told: The influence of the United States occupation on Japanese radio. In: *Radio Reader: Essays in the Cultural History of Radio*. Routledge, chapter 14.
- Strömberg, David, 2004. Radio's impact on public spending. *Q. J. Econ.* 119 (1), 189–221.
- Strömberg, David, 2015. Media coverage and political accountability. In: *Handbook of Media Economics*. pp. 595–622.
- Svensson, Jakob, Yanagizawa-Drott, David, 2009. Getting prices right: The impact of the market information service in uganda. *J. Eur. Econ. Assoc.* 7 (2–3), 435–445.
- The House of Representatives, 1947a. Minutes of the march 28, 1947 house of representatives committee meeting.
- The House of Representatives, 1947b. Minutes of the march 30, 1947 house of representatives floor meeting.
- Uemura, Chikako, 2007. Josei Kaiho O Meguru Senryo Seisaku [Occupation Policies on Women's Liberation]. Sosho jenda bunseki, Tokyo.
- Wang, Tianyi, 2021a. Media, pulpit, and populist persuasion: Evidence from father coughlin. *Am. Econ. Rev.* 111 (9), 3064–3092.
- Wang, Tianyi, 2021b. Waves of empowerment: Black radio and the civil rights movement.
- Yamamoto, Kiyoko, 2017. Senryoka Nihon ni okeru jinko yusei seisaku [population and eugenics policy in occupied Japan]. Sonoda Women's Coll. Stud..
- Yanagizawa-Drott, David, 2014. Propaganda and conflict: Evidence from the rwandan genocide. *Q. J. Econ.* 129 (4), 1947–1994.